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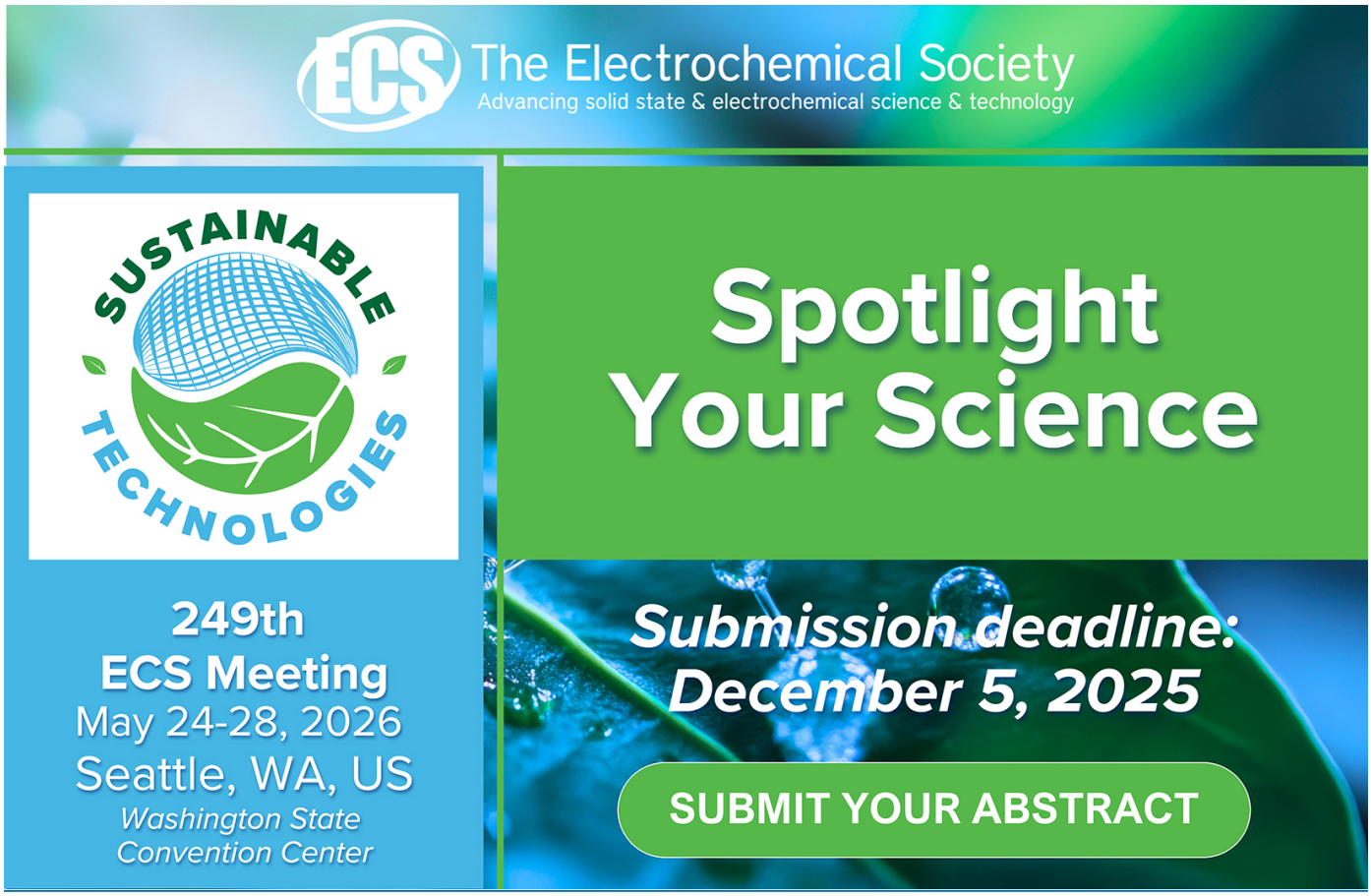

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Low-loss Optical Switching Based on Phase-change Material Sb_2Se_3 in a Micro-ring Resonator

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Abstract. In the past research, optical interconnection has successfully developed into an emerging technology due to its compatibility with CMOS, where an efficient and reconfigurable optical switch operating at ultra-low programming energy is a key element to realize optical signal routing and switching. Here we numerically demonstrate an optical switch operating in the telecommunications band, which shows a 20 dB switch ratio and insertion loss of 0.8 dB. The device is implemented in waveguide-coupled micro-ring resonator (MRR) using the phase materials Sb_2Se_3, which exhibits high contrast in its optical properties upon transitions between its crystalline and amorphous structural phases.

1. Introduction

Programmable silicon photonic integrated circuits (PICs) play a pivotal role in nanophotonic technology, and find extensive applications in fields such as fiber optics communications [1], spectroscopic sensors, optical neural networks [2], and quantum information processing. Notably, these applications often necessitate a multitude of optical components and switches. The advancement of PICs technology has accentuated the significance of compact size and energy efficiency in individual photon switches. Conventional optical switches typically rely on the injection of semiconductor carriers [3] and thermo-optic (TO) effect [4], or are formulated using volatile field-effect and limited complementary metal-oxide-semiconductor (CMOS) compatibility [5]. Unfortunately, due to the small refractive index, these mechanisms tend to yield sizable device footprints [6], ranging from several hundred micrometers and beyond, and a high switching energy consumption [7]. In contrast, phase-change materials (PCMs) offer promising solutions to address these challenges. Leveraging their non-volatile nature and high refractive index contrast, PCMs present feasible avenues for reducing device size and improving energy efficiency.

Among the various PCMs used in integrated photonics, $Ge_2Sb_2Te_5$ (GST) is the most extensively studied material to date. However, a major drawback is the high absorption loss in both phases within telecommunication C-band. The crystalline phase of GST exhibits an absorption loss (k) of 1.49, while the amorphous phase also shows significant absorption ($k = 0.12$) [8]. The high absorption loss is incompatible with integrated photonics platforms. To overcome this limitation, researchers have identified Sb_2Se_3, an antimony-based sulfur compound, as an ultra-low-loss phase-change material for PICs. Notably, both the crystalline and amorphous phases of Sb_2Se_3 exhibit negligible inherent

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absorption loss ($k < 10^{-5}$) at 1550nm [9]. Furthermore, Sb_2Se_3 maintains non-volatility and CMOS compatibility at a working temperature of 200°C [10], hence making Sb_2Se_3 highly promising for applications such as highly sensitive photodetector [11], electro-optic modulation, and sensing.

In this paper we propose a novel reconfigurable switch has the advantage of low insertion loss, high extinction ratio, and compact structure, which is based on phase-change materials Sb_2Sb_3. In our work, the area of 1.3×0.7 μm^2 Sb_2Sb_3 patch can be deposited between bus waveguide, acting as partially reflecting elements within the waveguide. We demonstrate that this structure can generate asymmetric Fano line shapes and achieve linear control by manipulating phase states and deposition areas. The numerical simulations results show that the feasibility of achieving a switch ratio as high as 20 dB due to the significant difference in refractive index [12] between amorphous Sb_2Se_3 (aSbSe) and crystalline Sb_2Se_3 (cSbSe), and the average insertion loss (IL) is around 0.8 dB.

2. Results and Discussion

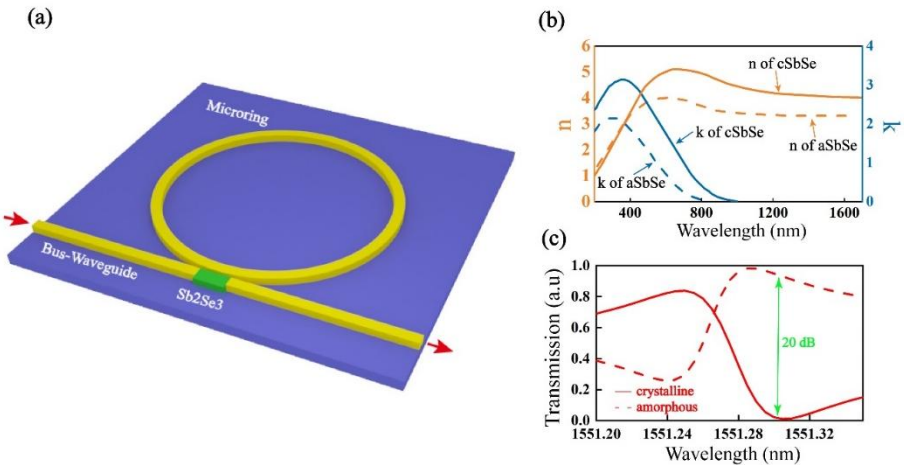

Figure 1. (a) The schematic illustration of the Sb_2Se_3 based optical switch. The structure consists of a Si waveguide layer integrated with Sb_2Se_3 film, and a silica substrate layer at the bottom. The length of Sb_2Se_3 patch is 1.3 μm, and the width is 700nm. (b) The real (yellow) and imaginary (blue) parts for the complex refractive index of aSbSe and cSbSe, respectively, at selected wavelength λ range from 200 to 1700nm. (c) The transmission spectra of this optical switch deposited Sb_2Se_3, showing almost complete on/off modulation of resonant peaks due to phase transition. The two spectra correspond to two states of Sb_2Se_3: (dashed) as-deposited amorphous state, (solid) crystalline state.

Figure 1(b) shows the real and imaginary parts of refractive index for PCM Sb_2Se_3 before and after phase transition, in the telecommunications C-band at 1550 nm, which indicates that n_{amorph} = 3.285 and n_{cryst} = 4.050, respectively[13]. With a bandgap energy of 1.2 eV, making the material absorption edge shifting into visible range without compromising the low absorption at telecommunication band. Hence, some commonly transitions triggered methods, such as thermal annealing[14], optical heating[13] and external heaters[15–17], also apply to Sb_2Se_3, enabling a large volume of Sb_2Se_3 to undergo a phase transition, allowing high extinction ratio photonic switches.

We next performed full-wave numerical simulations using a fully three-dimensional finite element technique (in COMSOL Multiphysics) to investigate the proposed optical switch based on PCM. The numerical simulations were conducted using. As illustrated in figure 1(a), the device is fabricated on a silicon-on-insulator chip, which has a 220 nm thick top silicon layer and a 2 μm thick buried dioxide layer. The bus waveguide and a Si resonator with radius of 10 μm have the same stripe width of 500 nm, and their coupling gap is 140 nm. Besides, the Sb_2Se_3 thin film is integrated within Si bus waveguide, locating at the middle of the coupling region with 1.3 μm (length)×700 nm (width). By

Micro-ring resonator

utilizing the design of a MRR coupled with a bus waveguide, we enhance the modulation effect of Sb_2Se_3. Figure 1(c) depicts simulated results of the Sb_2Se_3 based optical switch. To evaluate switching performance of this structure, we consider an important parameter of the modulator: the extinction ratio, which is calculated from the following equations:

$$Extinction\ ratio = 10 * \log_{10}\left[P(ON) / P(OFF)\right] \tag{1}$$

Here, *P(ON)* is the output signal while the switch is in "ON" state and *P(OFF)* is the output signal corresponding to the "OFF" state. The higher the extinction ratio becomes and the better performance the modulator has. We can easily find that a high-contrast optical switch with a 20 dB contrast ratio at the wavelength of 1551.3 nm, and the *P(ON)*, *P(OFF)* have been equal to 0.93 and 0.01 respectively. We designate amorphous phase as the "ON" state and crystalline phase as the "OFF" state. In this case, the insertion loss (IL) in the "ON" state is 0.8 dB.

We use the principle of interference to explain this structure. At the working wavelength position (1551.3nm), the phase difference between the light passing through the MRR and aSbSe is 0 and the two beams of light represent constructive interference, when there is light output in the output region of bus waveguide and the optical switch is in the "ON" state. At the same wavelength position, after Sb_2Se_3 phase transformation, the phase difference is π, leading to destructive interference with the two beams of light, and optical switch is in the "OFF" state.

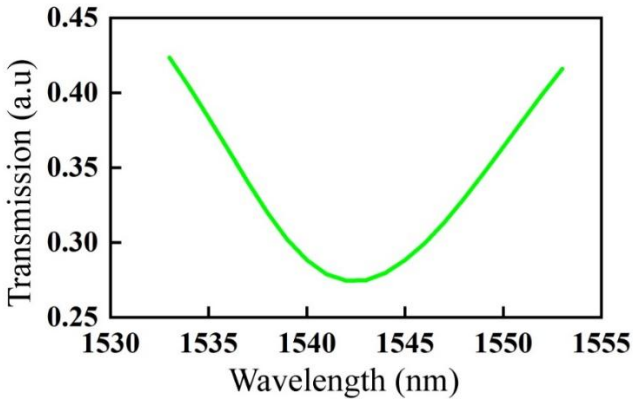

Figure 2. Transmittance spectra of a wide-spectrum background path.

In this side-coupled waveguide- resonator structure, as demonstrated in figure 1(a), the MRR and bus waveguide are indirect (resonant cavity) paths and background (direct) path [18], respectively, furthermore, the Sb_2Se_3-patch acting as partially reflecting elements within the bus waveguide constitute the Fabry-Perot resonator, which is crucial for the Fano line shape [19]. In absence of the resonant cavity coupling, the transmission of system is determined by the F-P resonator properties which show a wide-spectrum background path, as seen in figure 2. This wide-spectrum interferes with the narrow-band Lorentz line-shape of micro-ring cavity to form the Fano line.

Unlike the structure in the previous study which placed PCMs on the MRR disrupting the mode of the ring resonator and increasing transmission loss, we study that Sb_2Se_3 patch is placed between the bus waveguide, so that resonant wavelength will hardly move with phase transformation process. Moreover, Fano asymmetry is sensitive to the change of background path complex refractive index, and thus the linear shapes show opposite asymmetry when Sb_2Se_3 is in different states, that is why high-on/off-ratio can be achieved at the same operating wavelength, which is conducive to the miniaturization of the unit device. All-optical switch of Si-VO_2 using a straight waveguide or in a Mach-Zehnder interferometer or micro-ring resonator, also stands out as a promising device for photonics, which switching energies between 200 and 300 nJ were reported for this kind of device. However, optical modulation can only achieve 7 dB [20]. Recently, nonvolatile optical switching in a hybrid GST/Si waveguide device has been demonstrated by means of electrical excitation but heating

the silicon, which was highly doped to optimize the heat transfer process and achieve an ohmic contact with the metallic electrodes, but a high IL of 7.5 dB was reported owing to a non-amorphization of the whole active region [21].

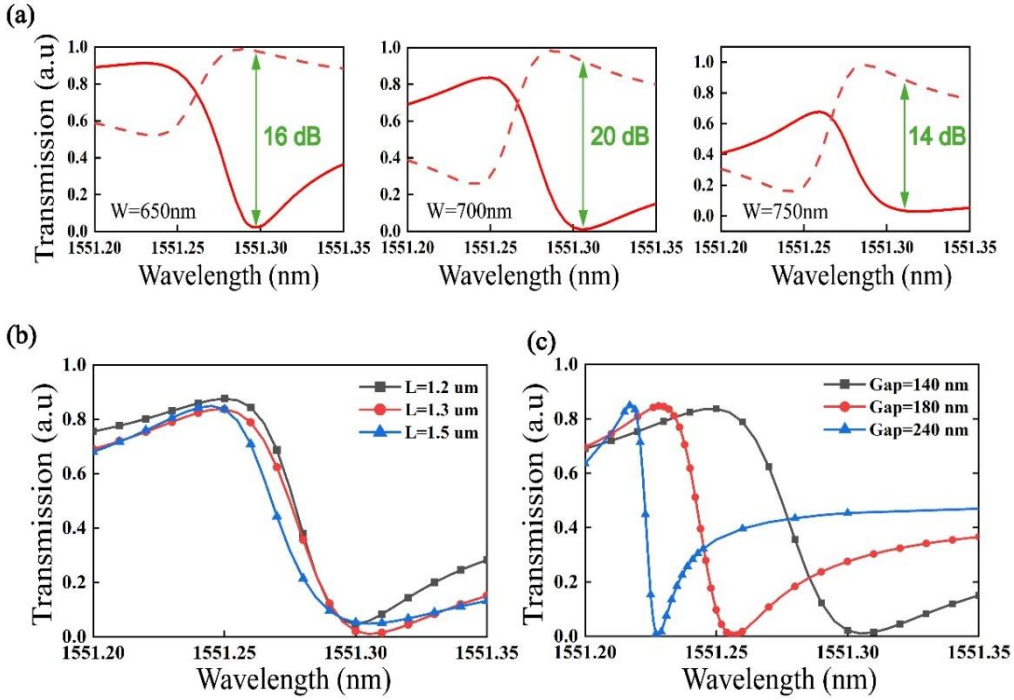

Figure 3. (a) Simulated transmission spectrum of the optical switch structure with different Sb_2Se_3 width, (b) length, and (c) the gap between resonant cavity and waveguide.

To achieve highest on/off ratio, we study the effect of some parameters, including the width and length of Sb_2Se_3 thin film. From figure 3(a) and (b), it is seen that the critical coupling can be achieved when the Sb_2Se_3 deposition length is 1.3 μm and the width is 700 nm, while the on/off ratio would be decreased in other deposition sizes, according to Eq. (1), this is because the *P(OFF)* equal to 0.023, 0.01, 0.03 respectively when *W* varies from 650 to 750 nm, and equal to 0.046, 0.01, 0.05 when *L* varies from 1.2 to 1.5 μm. The insertion loss in "ON" state will increase with the increase of Sb_2Se_3 width. Thus, the Sb_2Se_3 geometry optimization is crucial for increasing the modulation depth and reducing IL. Considering the experimental processing, we also analyze the influence of the gap between resonant cavity and background path. Shown in Figure 3(c), it is seen that the gap variation has little effect on the performance of this switch, and even when gap is increased to 240nm, the resonance has a higher Q value and can reach a high on/off ratio of 27dB. Therefore, the gap value can be appropriately increased during experimental processing.

In this regard, it is worth mentioning that although early works had shown relatively slow amorphous to crystalline switching times and physically limited to hundreds of picoseconds [22], crystallization of Sb_2Se_3 with 0.1-1 ms pulses has been recently experimentally demonstrated. Besides, the operation speed depends on a variety of factors including the thermal design, film dimensions, and method of triggering the phase transition. Improving the timescale for the transition could be achieved through more sophisticated thermal design, such as reducing the Sb_2Se_3 thickness, depositing a top-cladding (e.g. SiO_2), or tailoring the SOI substrate geometry.

3. Conclusions

In conclusion, we propose an optical switch based on phase-change materials Sb_2Sb_3 and numerically demonstrate that it has both low insertion loss, high extinction ratio, and compact structure. We use the principle of interference to explain the cause of Fano line shape, and we analyze the influence of the Sb_2Sb_3 length, width to achieve high on/off ratio. Using numerical simulations, we have achieved a high on/off ratio of 20 dB and insertion loss in the "ON" state of 0.8 dB, operating at telecommunication wavelengths. These findings highlight the potential of Sb_2Se_3 as a superior PCM for integrated photonics, addressing the issue of high absorption loss in previous materials. Further research and development in this area hold promise for realizing high-performance photonic devices with minimal power consumption and increased functionality.

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